



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

SEPTEMBER 2008
YEARLY EXAMINATION
YEAR 11 Continuers

Mathematics Preliminary

General Instructions:

- Working time—90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.
- Answer in simplest exact form unless otherwise instructed.

Total marks—80 Marks

- Attempt all questions.
- Start each NEW section in a separate answer booklet.
- Hand in your answer booklets in 4 sections: A, B, C, D.

Examiner: Mr D. Hespe

Section A — Start a new booklet

Marks

Question 1 (20 marks)

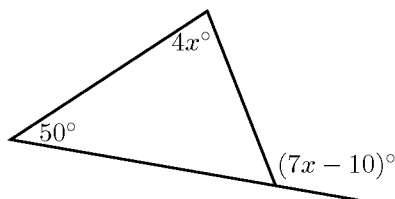
- (a) Evaluate $\frac{(3.26)^2}{16.39 - 12.51}$ correct to two decimal places. 2
- (b) Express 0.00035 in scientific notation. 1
- (c) (i) Find the reciprocal of $2\frac{3}{4}$. Answer as a fraction. 1
(ii) Also express the answer as an exact decimal. 1
- (d) Simplify $\sqrt{147} - \sqrt{75}$. 2
- (e) Evaluate $\cot 3$ correct to 3 significant figures. 2
- (f) Express $\frac{2\sqrt{3}}{\sqrt{3} - 1}$ with a rational denominator in simplest form. 2
- (g) Factorise $2x^2 + 3x - 2$. 2
- (h) Solve $5t + 3 = 2(1 - t)$. 2
- (i) Find the slope of the normal to the line $2x - 3y + 6 = 0$. 2
- (j) Sketch the curve $y = \sqrt{4 - x}$ for $0 \leq x \leq 4$. 3

Section B — Start a new booklet

Question 2 (20 marks)

(a) If $r = \sqrt{\frac{A}{4\pi}}$, find A (to the nearest integer) when $r = 6.25$. 2

(b) Find the value of x , giving reasons. 4

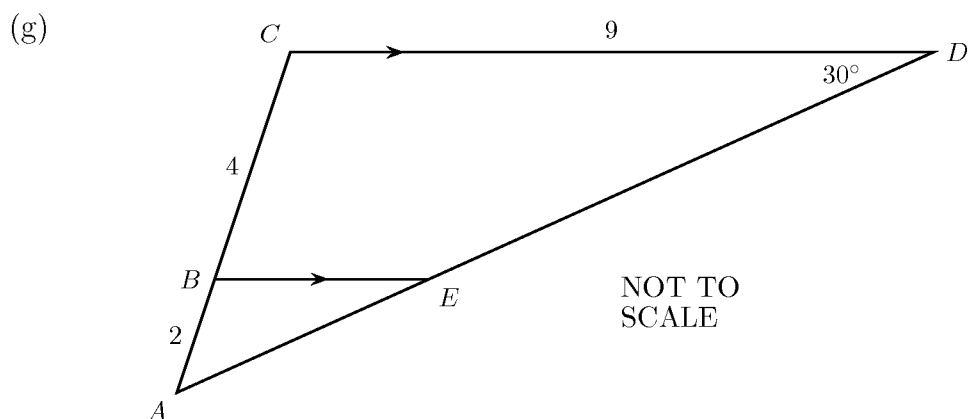


(c) Solve the pair of equations $3x - 2y = 9$ and $x + 4y = -4$. 3

(d) Factorise completely $x^4 - xy^3 + x^3y - y^4$. 2

(e) Write down the domain and range of the function $y = 2 + |x - 3|$. 2

(f) A triangle is isosceles with its vertex angle 20° more than a base angle. Find the measure of the base angle. 2



In the diagram, ACD is a triangle where $AB = 2$ cm, $BC = 4$ cm, $CD = 9$ cm, and $\angle CDE = 30^\circ$. Also, BE is parallel to CD .

(i) Find the size of $\angle BED$. Give a reason for your answer. 1

(ii) Find the length of BE . Give reasons for your answer. 4

Section C — Start a new booklet

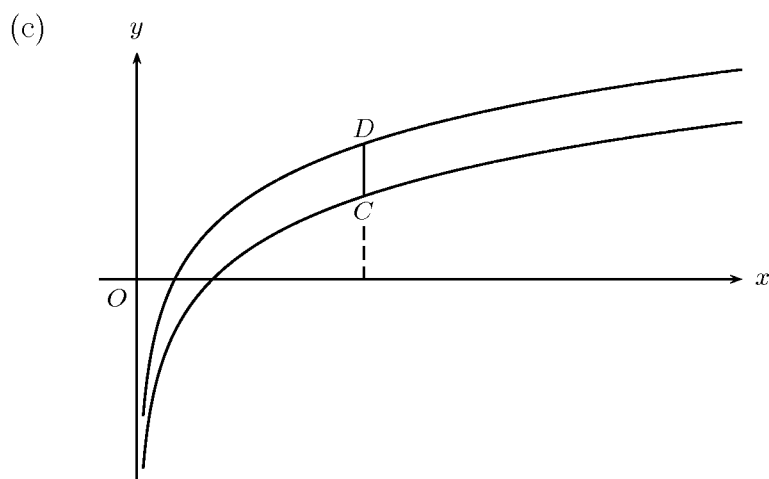
Question 3 (20 marks)

(a) Solve for x : $\log_7 \left(\frac{x-4}{x-1} \right) = 2$.

2

(b) Find the equation in general form of the line through the point $(2, 0)$ and the intersection of the two lines $x + y + 1 = 0$, $2x - y = 3$.

3



2

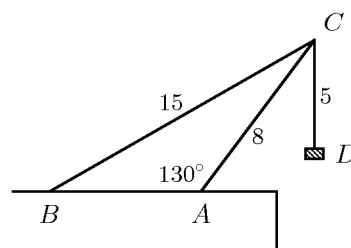
A vertical line is drawn to cross the two graphs $y = \log_{10} x$ and $y = \log_{10} 2x$ at points C and D . Show that the distance CD is constant (that is, does not depend on the position where the vertical line is drawn).

(d) (i) In the shear-legs shown to the right, find the height of the load D above the horizontal level of AB . Note that all lengths are in metres and answer to 2 decimal places.

2

(ii) Find also the length of AB correct to 2 decimal places.

3



(e) Consider the parabola $12y = x^2 - 6x - 3$.

2

(i) Find the equation of its directrix.

(ii) Find the coordinates of its focus.

2

(f) Find all real numbers x which satisfy the equation $4x^4 = 4x^2 + 3$.

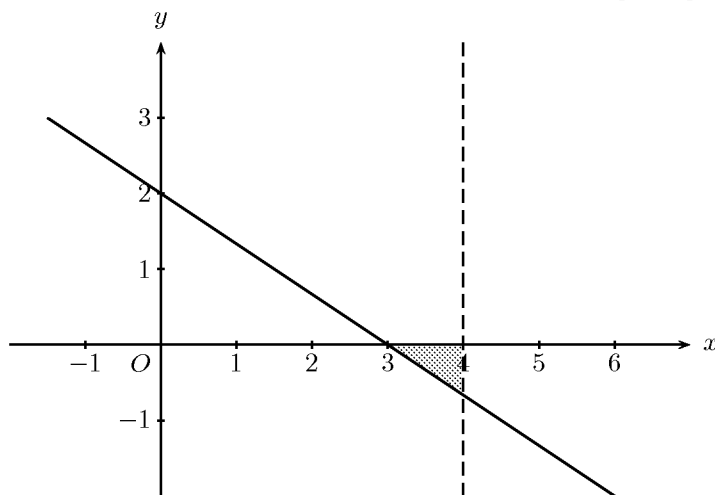
4

Section D — Start a new booklet

Question 4 (20 marks)

- (a) Write down three inequalities to describe the shaded region given.

3



- (b) The coordinates $A(-1, 1)$, $B(1, 4)$, and $C(10, -2)$ are joined to form $\triangle ABC$.

(i) Draw a diagram on a number plane, marking on it the information supplied.

1

(ii) Prove that $\triangle ABC$ is right-angled.

2

(iii) Find the coordinates of M , the midpoint of the hypotenuse.

1

(iv) Show that M is the centre of a circle that could be drawn through A , B , and C .

3

- (c) A triangle PQR has side lengths $p = 20$, $q = 15$, $r = 11$.

(i) Find the size of each angle.

4

(ii) Hence or otherwise, find its area.

2

- (d) Prove that $\sin^2 x = 1 - \frac{1}{1 - \frac{1}{1 - \frac{1}{\sin^2 x}}}$

4

End of Paper

Question 1

$$a) (3.26)^2$$

$$16.39 - 12.51$$

$$= 2.739072165$$

$$= 2.74 \text{ (2dp)} \quad (1)$$

$$b) 0.00035$$

$$= 3.5 \times 10^{-4} \quad (1)$$

$$c) i) 2^{3/4} = 11/4$$

$$\text{reciprocal} = 4/11 \quad (1)$$

$$ii) 4/11 = 0.36 \quad (1)$$

$$d) \sqrt{147} - \sqrt{75}$$

$$= \sqrt{49} \times \sqrt{3} - \sqrt{25} \times \sqrt{3} \quad (1)$$

$$= 7\sqrt{3} - 5\sqrt{3}$$

$$= 2\sqrt{3} \quad (1)$$

$$e) \cot 3 = \frac{1}{\tan 3} \quad (1)$$

$$= 19.08113669$$

$$= 19.1 \text{ (3 sig fig)} \quad (1)$$

$$f) \frac{2\sqrt{3}}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1}$$

$$= \frac{6+2\sqrt{3}}{3-1} \quad \begin{array}{l} -1 \text{ for 1st} \\ \text{mistake} \end{array}$$

$$= \frac{6+2\sqrt{3}}{2} \quad \begin{array}{l} 1/2 \text{ for each} \\ \text{extra.} \end{array}$$

$$= 3+\sqrt{3} \quad (2)$$

$$g) 2x^2 + 3x - 2^{-4}$$

$$= (2x+4)(2x-1)$$

$$= \frac{2(x+2)(2x-1)}{2}$$

$$= (x+2)(2x-1) \quad (2)$$

$$h) 5t+3=2(1-t)$$

$$5t+3=2-2t$$

$$7t = -1$$

$$t = -1/7$$

$$i) 2x-3y+6=0$$

$$3y = 2x+6$$

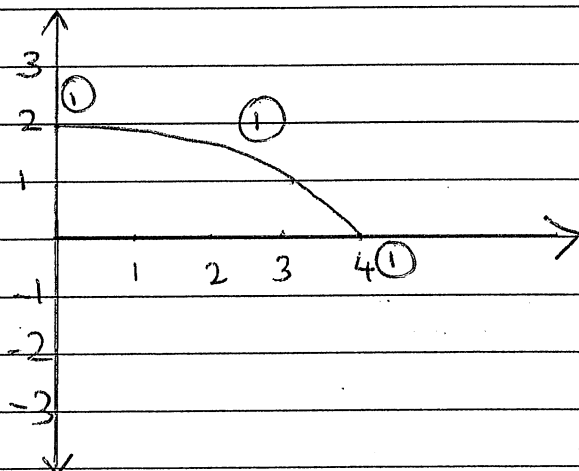
$$y = 2/3x + 2$$

$$\therefore \text{slope} = 2/3 \quad (1)$$

$$\therefore \text{slope of normal} = -3/2 \quad (1)$$

$$j) y = \sqrt{4-x} \text{ for } 0 \leq x \leq 4$$

x	0	1	2	3	4
y	2	$\sqrt{3}$	$\sqrt{2}$	1	0



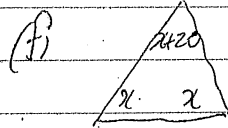
SECTION B

$$(a) r = \sqrt{\frac{A}{4\pi}} = 6.25$$

$$\frac{A}{4\pi} = 6.25^2$$

$$A = 6.25^2 \times 4\pi = 490.87$$

491 nearest integer



$$3x + 20 = 180 \text{ (}\angle \text{ sum of } \triangle \text{)}$$

$$3x = 160$$

$$x = 53\frac{1}{3}^\circ = \text{base angle}$$

$$(b) 4x + 50 = 7x - 10 \text{ (Ext. } \angle = \text{ sum of int. op. angles)}$$

$$60 = 3x$$

$$x = 20$$

$$(g) \angle BED = 180 - 30 = 150^\circ$$

(i) (Count. to $\angle CDE$)

$$(ii) \text{ In } \triangle ACD + \triangle ABE$$

$\angle CAD$ is common

$$\angle BEA = \angle CDE$$

(Corr. $\angle$ s $CE \parallel BE$)

$\therefore \triangle ACD \parallel \triangle ABE$ (equiangular)

$$\therefore \frac{BE}{9} = \frac{2}{6} \text{ (Corr sides)}$$

$$BE = 3 \text{ u}$$

$$(d) x^4 - xy^3 + x^3y - y^4$$

$$x(x^3 - y^3) + y(x^3 - y^3)$$

$$(x+y)x(x^2 - y^2)$$

$$(x+y)x(x-y)(x^2 + xy + y^2)$$

$$(e) \text{ Domain: All real } x$$

$$\text{Range: } y \geq 2$$

Q (3) (a) $\frac{x-4}{x-1} = 7^2$

$$x-4 = 49(x-1)$$

$$x-4 = 49x-49$$

$$-4+49 = 48x$$

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$$45 = 48x$$

$$x = \frac{15}{16}$$

(2)

OR.

$$(x+y+1)+k(2x-y-3)=0$$

$$(2,0) (2+0+1)+k(4-0-3)=0$$

$$3+k=0$$

$$k=-3$$

(b) $x+y=-1$
 $2x-y=3$ +

$$3x = 2$$

$$x = \frac{2}{3}$$

$$y = -1\frac{2}{3}$$

$$\left(\frac{2}{3}, -1\frac{2}{3}\right) (2,0)$$

$$m = \frac{0+1\frac{2}{3}}{2-\frac{2}{3}} = \frac{1\frac{2}{3}}{1\frac{1}{3}} = 1\frac{1}{4} = \frac{5}{4}$$

$$(y-0) = \frac{5}{4}(x-2)$$

$$5x-4y-10=0$$

(3)

$$(c) \quad y = \log_{10} x \quad y = \log_{10} 2x.$$

At $x=x$, $(x, \log_{10} x)$ and $(x, \log_{10} 2x)$.

$$\begin{aligned} d &= \sqrt{(x-x)^2 + (\log_{10} 2x - \log_{10} x)^2} \\ &= \log_{10} 2x - \log_{10} x \\ &= \log_{10} 2 + \log_{10} x - \log_{10} x \\ &= \log_{10} 2. \quad (2) \end{aligned}$$

$$(e) \quad 12y = x^2 - 6x - 3.$$

$$12y + 3 = x^2 - 6x$$

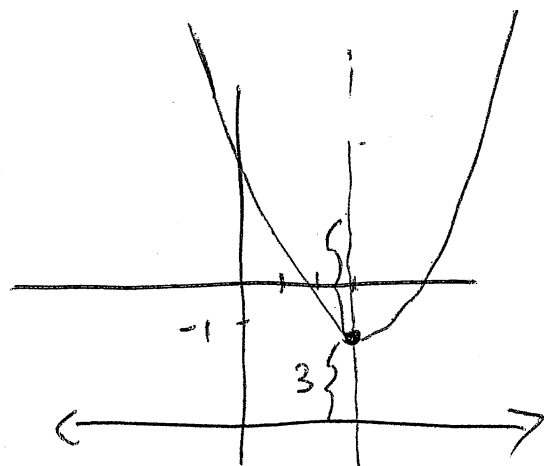
$$12y + 3 + 9 = x^2 - 6x + 9$$

$$12y + 12 = (x-3)^2$$

$$(x-3)^2 = 12(y+1)$$

$$(x-h)^2 = 4a(y-k)$$

$$a=3, V(h,k) = (3, -1)$$



$$(i) \quad y = -4 \quad (2)$$

$$(ii) \quad S(3, 2). \quad (2)$$

$$(P) \quad 4x^4 = 4x^2 + 3$$

$$4x^4 - 4x^2 - 3 = 0$$

$$\text{let } u = x^2, \quad 4u^2 - 4u - 3 = 0$$

$$(2u + 1)(2u - 3) = 0$$

$$u = -\frac{1}{2} \text{ and } u = \frac{3}{2}.$$

So $x^2 = -\frac{1}{2}$.
no real solns.

$$x^2 = \frac{3}{2}$$

$$x = \pm \sqrt{\frac{3}{2}}$$

(4)

SECTION D

Question 4 (3)

(a) Eqⁿ of oblique line

$$\frac{x}{3} + \frac{y}{2} = 1$$

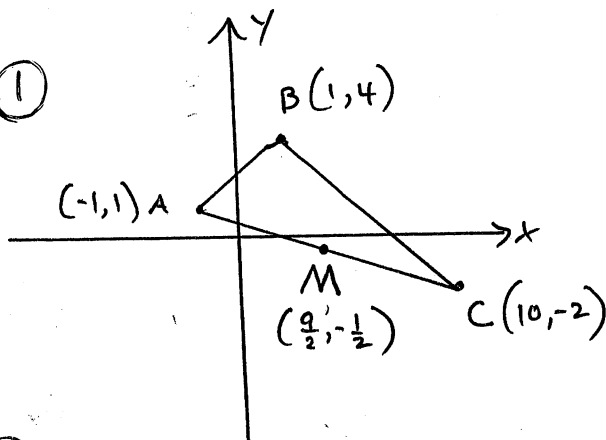
$$\text{i.e. } 2x + 3y = 6$$

$\therefore$ Region defined by:

$$2x + 3y \leq 6 \text{ and } x < 4 \\ \text{and } y \geq 0$$

(b)

(i) ①



$$(ii) \text{ ② } m_{AB} = \frac{4-1}{1-(-1)} = \frac{3}{2}$$

$$m_{BC} = \frac{-2-4}{10-1} = -\frac{2}{3}$$

$$\text{Since } m_{AB} \times m_{BC} = -1$$

$$\Rightarrow AB \perp BC$$

$$\therefore \angle ABC = 90^\circ$$

(or use Pyth. Th.) and show

$$(AC)^2 = (AB)^2 + (BC)^2$$

① (b) (iii) ctd

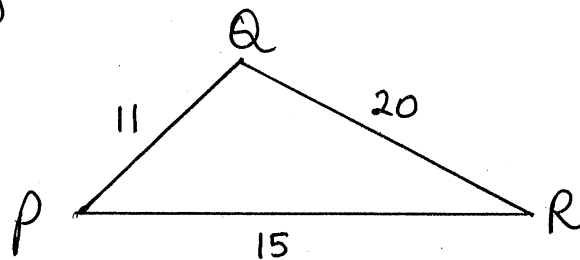
$$M\left(\frac{10+(-1)}{2}, \frac{-2+1}{2}\right) = \left(\frac{9}{2}, -\frac{1}{2}\right)$$

(iv) ③ If $AM = BM = MC \Rightarrow A, B, C$ lie on circle with above equal radii, centre M .

$$AM = \sqrt{\left(\frac{9}{2} - (-1)\right)^2 + \left(-\frac{1}{2} - 1\right)^2} = \sqrt{\frac{130}{4}} = MC$$

$$BM = \sqrt{\left(\frac{9}{2} - 1\right)^2 + \left(-\frac{1}{2} - 4\right)^2} = \sqrt{\frac{130}{4}}$$

(c) ④ (i)



$$\cos P = \frac{11^2 + 15^2 - 20^2}{2 \cdot 11 \cdot 15} \Rightarrow P = 99^\circ 25'$$

$$\cos Q = \frac{11^2 + 20^2 - 15^2}{2 \cdot 11 \cdot 20} \Rightarrow Q = 47^\circ 43'$$

$$\therefore R = 32^\circ 52'$$

(ii) ②

$$\text{Area } \triangle PQR = \frac{1}{2} \cdot 11 \cdot 20 \sin Q \\ = 81.38 \text{ units}^2$$

(d) (4)

$$\text{RHS} = 1 - \frac{1}{1 - \frac{1}{1 - \frac{1}{S^2}}}$$

$$S = \sin x$$

$$= 1 - \frac{1}{1 - \frac{1}{1 - \csc^2 x}}$$

$$\text{Since} \\ 1 + \cot^2 x = \csc^2 x$$

$$= 1 - \frac{1}{1 - \frac{1}{-\cot^2 x}}$$

$$= 1 - \frac{1}{1 + \tan^2 x}$$

$$= 1 - \frac{1}{\sec^2 x}$$

$$\text{since} \\ 1 + \tan^2 x = \sec^2 x$$

$$= 1 - \cos^2 x$$

$$= \sin^2 x$$

$$\text{Since} \\ \sin^2 x + \cos^2 x = 1$$

$$= \text{LHS}$$